Technical Syllabus for a 16-Year Java, Microservices, Azure DevOps Professional

Roadmap to remain competitive with modern cloud-native, platform engineering, and AI-enabled software developers.

1. Core Java Mastery

- Java 17/21 features
- JVM internals
- Garbage collection
- Concurrency
- CompletableFuture
- Performance tuning

2. Spring Boot & Microservices

- Advanced Spring Boot
- Spring Cloud
- Saga/CQRS
- Distributed systems
- Resilience patterns

3. API Architecture

- REST
- OpenAPI
- GraphQL
- gRPC
- API Gateways

4. Databases

- SQL optimization
- PostgreSQL
- Cosmos DB
- MongoDB
- Redis

5. Azure Cloud

- AZ-104
- AZ-305
- AKS
- Networking
- Security

• Entra ID

6. Azure DevOps & Platform Engineering

• YAML pipelines
• GitOps
• Terraform
• Bicep
• Backstage

7. Kubernetes

• Pods
• Ingress
• Helm
• KEDA
• Service Mesh

8. Event Driven Architecture

• Kafka
• Azure Service Bus
• Event Grid
• Event Hub

9. Observability & SRE

• Prometheus
• Grafana
• OpenTelemetry
• SLI/SLO/SLA

10. Security

• OWASP Top 10
• DevSecOps
• Secrets Management

11. System Design

• Scalability
• High Availability
• Disaster Recovery
• Architecture Diagrams

12. AI Fundamentals

• LLMs

• Embeddings
• Vector Databases
• Prompt Engineering

13. AI Development

• Azure OpenAI
• AI Foundry
• Semantic Kernel
• LangChain

14. AI Agents

• LangGraph
• AutoGen
• CrewAI
• Tool Calling

15. AI Assisted Development

• GitHub Copilot
• Agentic Coding
• AI-generated tests

16. Testing

• JUnit 5
• Mockito
• Playwright
• JMeter
• K6

17. Principal Engineer Skills

• ADR
• FinOps
• Technology Roadmaps
• Developer Experience

12-Month Learning Plan

Quarter 1: Java 21, Spring Boot Advanced, Kubernetes Basics, Terraform, Azure Architecture

Quarter 2: Kafka, AKS, Prometheus, Grafana, OpenTelemetry

Quarter 3: AI Fundamentals, Azure OpenAI, RAG, Semantic Kernel, AI Agents

Quarter 4: System Design, Platform Engineering, Security, FinOps, Principal Engineer Skills

Top 5 Focus Areas: Kubernetes & AKS, Platform Engineering, Observability, AI Engineering, and Large Scale System Design.